

100 Days of Azure DevOps

Day 26 Handout — Pin the interpreter, then make ruff and pytest run on the agent — not on your laptop

3 - Continuous Integration · Personal lab only · Educational content · Not a sales pitch

Architecture A — Python CI is a contract with the hosted agent

Piece	YAML / file	What it proves
Interpreter	UsePythonVersion@0 versionSpec: '3.11'	Not leftover python3 on ubuntu-latest
Lockfile	requirements.txt (pytest, ruff pinned)	Next clone does not hunt a wiki
Lint	ruff check src/sample-python	Style + a class of bugs; own exit code
Tests	pytest -q --junitxml=\$(Build.StagingDirectory)/junit.xml	Behavior. Do not merge with lint
Receipt	PublishTestResults@2 on that junit.xml	The run has evidence, not only a green job

Architecture B — decide what fails the job

Signal	Fail the job when	Lab exception (temporary)
Interpreter missing	UsePythonVersion@0 cannot resolve 3.11	Never. Pin or the pipeline is folklore
Import / dep	pip install exits non-zero	Never. A missing wheel is a real break
ruff	ruff check finds F/E rules you adopted	true once so the first YAML exists
pytest	any test fails or collects 0 tests you expected	Never. Zero tests is not a safety net

Architecture C — pitfalls that go green for the wrong reason

Pitfall	What you see	Fix
Agent Python	Works locally, ImportError on ubuntu-latest	UsePythonVersion@0; echo \$(python --version)
No requirements.txt	pip install pytest ruff typed from memory	Commit the pins; Cache@2 on pip cache later
true forever	ruff red in the log, job still green	Remove the swallow before main is protected
pytest not collected	pytest -q collected 0 items → still 0	Put tests under tests/ or test_*.py; fail if empty

One-liner to remember

pytest is the friend who tells you the truth before your users do — but only if CI actually runs it.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal Azure DevOps org only. Project azure-100-labs. Hosted pool: vmImage ubuntu-latest.

- 1 In the lab repo create src/sample-python/app.py (a tiny function) and tests/test_app.py with one pytest that asserts it.
- 2 Add src/sample-python/requirements.txt with pytest==8.* and ruff pinned. Commit on a feature branch — do not put secrets in the file.
- 3 Pipelines → New pipeline → Azure Repos Git → azure-100-labs. Point at pipelines/python-ci.yml (starter below).
- 4 Run once. Open the job log: confirm UsePythonVersion printed 3.11, ruff ran, pytest collected ≥1 item.
- 5 If ruff is noisy, keep || true for this lab only and write that debt in docs/python-ci-day26.md. Do not leave it unexplained.
- 6 Optional: add PublishTestResults@2. Confirm the Tests tab on the run shows the case. Screenshot no secrets.

Starter YAML — pin 3.11, lint, pytest

```
# pipelines/python-ci.yml
trigger:
  branches:
    include: [ main ]
  paths:
    include: [ src/sample-python/**, pipelines/python-ci.yml ]
pool:
  vmImage: ubuntu-latest
steps:
- task: UsePythonVersion@0
  inputs:
    versionSpec: '3.11'
    addToPath: true
- script: |
  python -m pip install --upgrade pip
  pip install -r src/sample-python/requirements.txt
  ruff check src/sample-python || true
  pytest -q src/sample-python tests -q
  displayName: pip, ruff, pytest
```

Done checklist

- UsePythonVersion@0 pinned 3.11 in the log — not 'whatever the image had'
- At least one pytest collected and passed on the agent
- requirements.txt is committed; no passwords in YAML
- Posted the LinkedIn document (personal account, no employer)

Tomorrow — Day 27

CI for Java/Maven — JavaToolInstaller@0, the Maven lifecycle train, and mvn -B test.